

Are engineers ready to make the jump from knobs to a smartphone/tablet interface where they use their fingers to zoom in on a part of a waveform?

- "Manipulating the image on an oscilloscope by using touch-screen technology is a relatively easy jump for most engineers to make. A more difficult transition would be to use this kind of interface as the main screen to configure all the available functions in an oscilloscope with additional functions integrated."—**Pamela Aparo, marketing manager-electronic test & measurement, Analog Devices Inc.**
- "The new generation of engineers is smartphone savvy and enjoys working with the scope like an electronic gadget. Hence the user interface of the scopes is moving towards touch-screen operations."—**Srinivasa Appala, manager-product support & application, Rohde & Schwarz**
- "The proliferation of consumer electronics with capacitive touch-screens and intuitive operation is creating a new set of user expectations that, as a test and measurement vendor, we are continuing to track and evaluate."—**David Farrell, general manager-mainstream oscilloscopes, Tektronix**
- "Yes. If a common man can zoom his images and watch closely on smartphones then engineers would surely appreciate and use it with warm welcome."—**Vivek Mantri, country manager-industrial segment, Scientech Technologies Pvt Ltd**

beginning of 2000, informs Sharma. He says, "There has been a growing need for detecting digital signals which are usually presented by two discrete voltage levels." A logic analyser, which is the best fit for such digital-signal measurements, has the benefit of multiple-channel input measurements, usually limited to two or four channels in oscilloscopes. He adds, "Nowadays, some system designers not only need an MSO but also an instrument which can afford to do frequency- and time-domain analysis simultaneously. So, a mixed-domain oscilloscope (MDO) is the latest trend."

The continuously-evolving integration level of embedded designs drives further feature integration into oscilloscopes. Appala says, "Additional digital channels, as part of mixed-signal options, protocol-specific trigger and decode options are a few examples. The key requirement is a synchronised operation of such features for true system-level debugging."

In the analogue to digital era, a significant number of digital designs came onto the analogue board, thus MSOs came to light about ten years ago. "The trend has evolved since, and mixed-domain oscilloscopes came into existence with Tektronix leading the charge. Today, mixed domain has moved to another level with the inclusion of wireless RF signals," says David Farrell, general manager-mainstream oscillo-

scopes, Tektronix. He adds, "Analogue, digital and now RF signals are coming together to the embedded domain and creating significant changes on how work is done in designs houses and research centres, specifically with respect to analysing those signals in as short a time as possible."

On a similar note, Aparo says, "Mixed-domain oscilloscopes offer similar benefits as mixed-signal oscilloscopes when an engineer needs to analyse signals in both the time domain and frequency domain—and these are useful in looking for problems that are caused by phase instability or phase offsets."

Not everyone is into MDOs though. On the contrary, Sanchit Bhatia, digital applications specialist, Agilent Technologies India believes, MDOs are an attempt to add spectrum analyser capability to oscilloscopes. He says, "MSOs have not evolved to MDOs as yet due to lower specifications. In fact, in MDOs, the oscilloscope and spectrum analyser capabilities have gone backwards. Any general oscilloscope and any entry-level spectrum analyser have better specifications than today's MDOs." Supporting his view, he adds, "A stand-alone spectrum analyser and a stand-alone oscilloscope are a better solution than an MDO, as they offer full instrument capabilities and can be synchronised using external trigger, if the application so demands."

Our fast-moving world needs to analyse data fast, and also on the move, informs Vivek Mantri, country manager-industrial segment, Scientech Technologies. He says, "Trending technology in telecom and RF segments needs data analysis to be extremely accurate in time to display waveforms, and in frequency to display signal spectra, where MDOs are a perfect choice." However, he adds, "For most of the general-purpose requirements, analogue oscilloscopes are sufficient."

Handheld, PC-based, phone and even a wristwatch scope

Oscilloscopes have evolved from the typical bench-top instruments to as small as a wristwatch, providing flexibility and an array of choices. Mantri defines today's generation as tech-savvy, that is more comfortable with touch screens and USB data transfers. He says, "They want their oscilloscopes to be handheld, with an easy interface, and prefer data transfers through SD cards. Different user segments will find analogue, MSO or PC-based instruments useful, depending on their usage." Although, he maintains, no single technology can be suitable for all kinds of applications.

Today's handheld oscilloscopes range from 20MHz to 500MHz in bandwidth, with sampling rates from 500MSa/s to 5GSa/s, and mostly come with LCD displays. Typically, with two channels, these are ideally suited for use by hobbyists, students, technicians and anyone else looking for an easy-to-use, affordable, single-channel PC oscilloscope.

PC-based oscilloscopes, complemented with software, provide bandwidth signal analysis anywhere between few tens of MHz to as high as 20GHz. These USB, modular devices typically have one, two, four or even eight channels with USB 2.0 or USB 3.0 interface, depending on the model. Some of them are even loaded with memory capacity. The software provided with these instruments is used to transfer instrument set ups, meas-